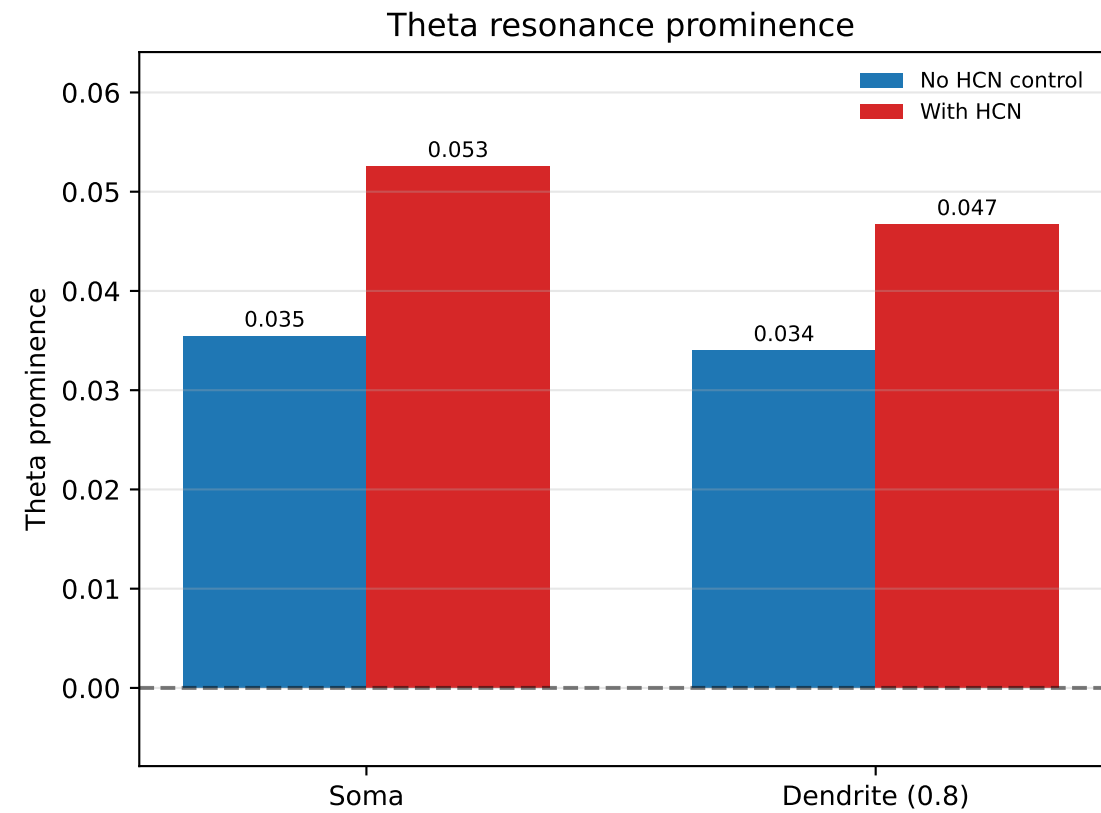
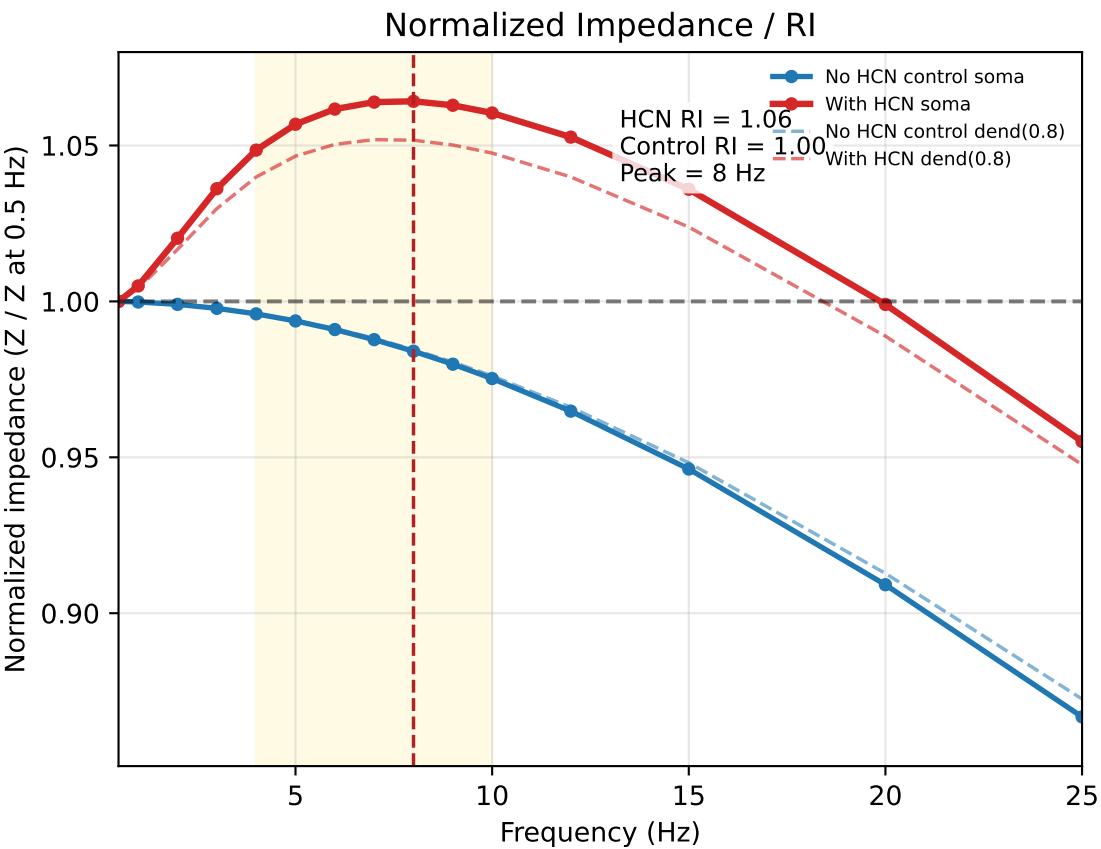
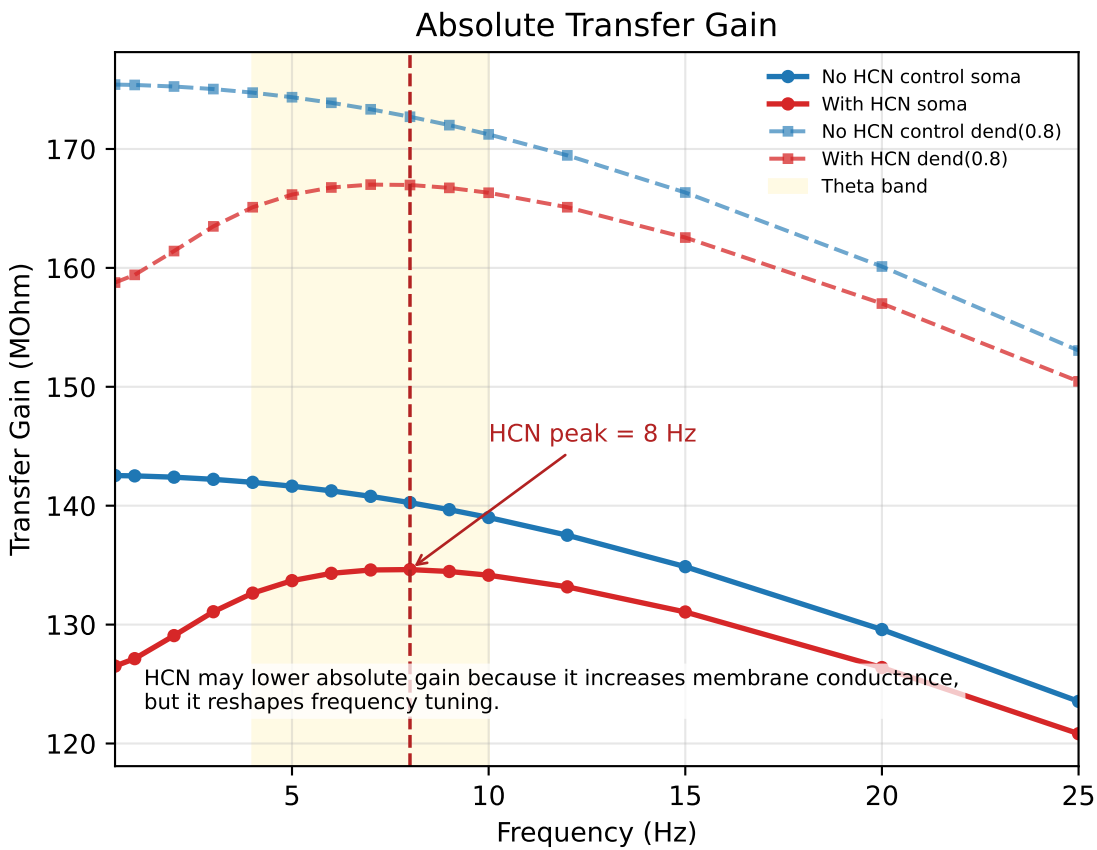
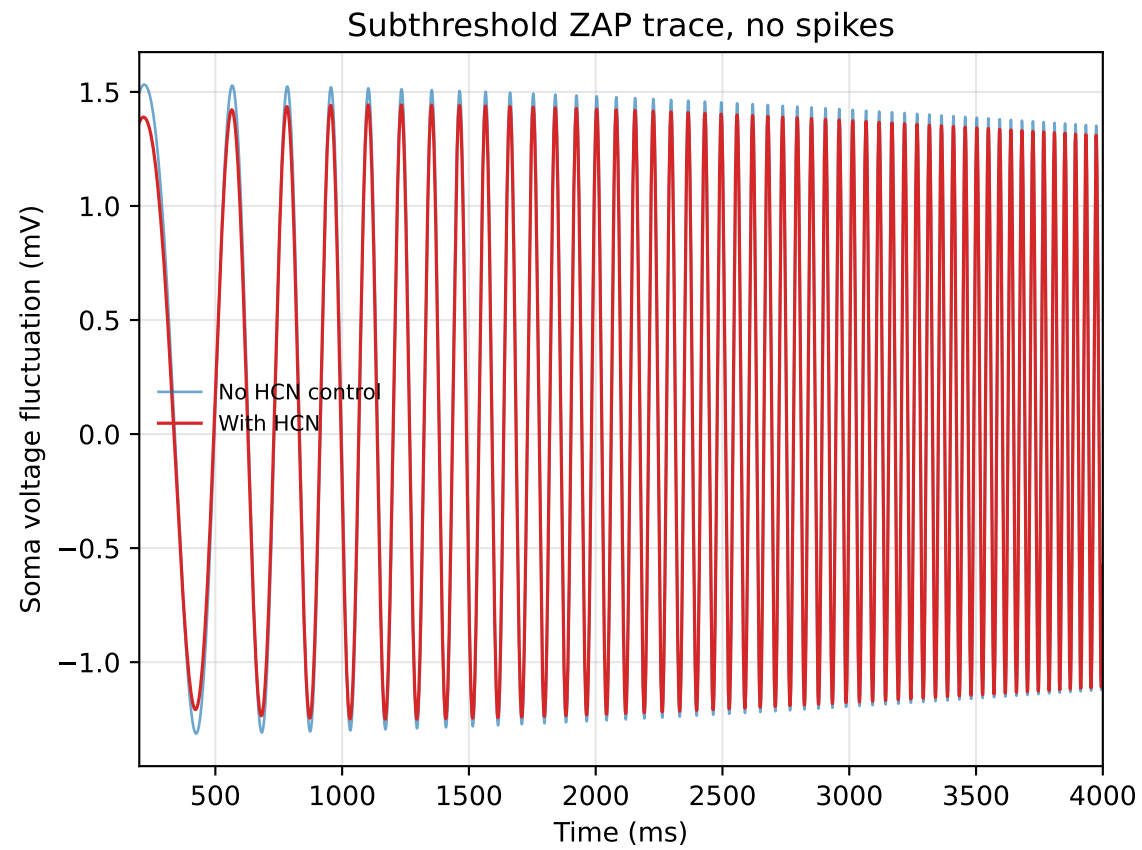
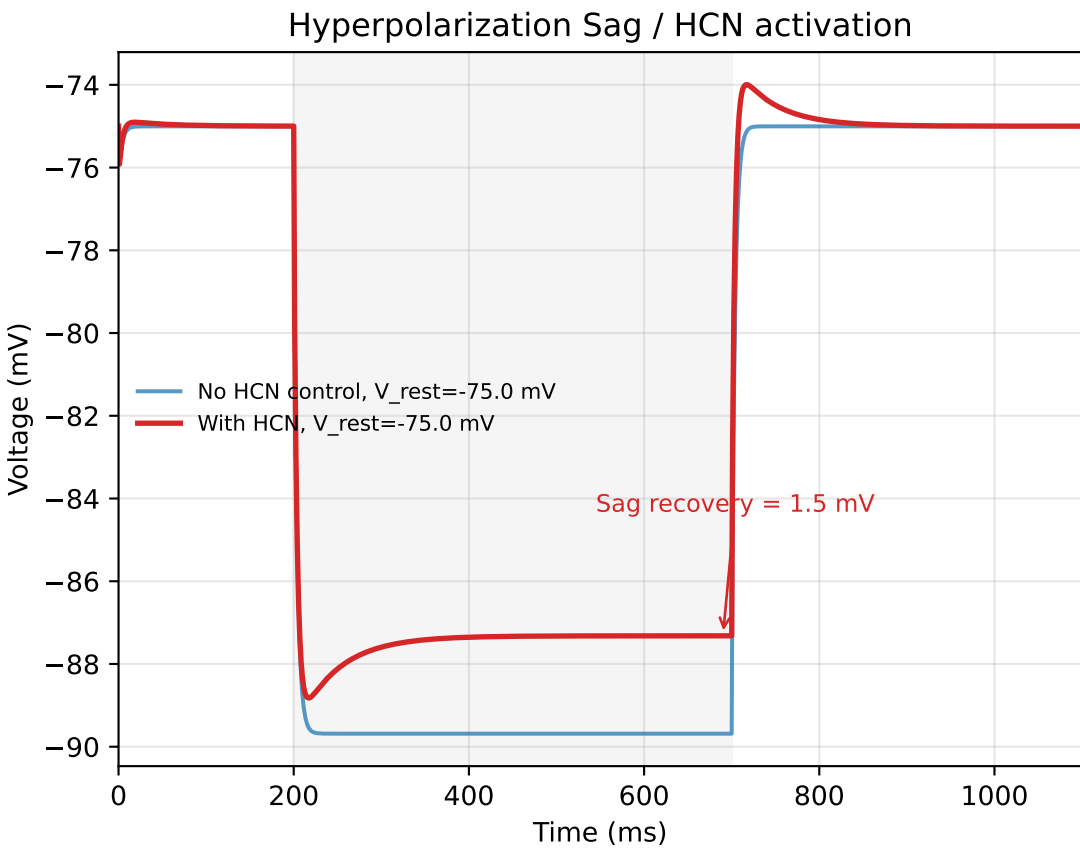


HCN/Ih reshapes subthreshold membrane frequency tuning toward theta-band resonance



Model:

- No HCN control: pas + somatic HH/HH1
- With HCN: pas + somatic HH/HH1 + Ih
- Input: distal dendritic current injection at dend(0.8)
- Recording: soma and dend(0.8)
- Holding target: -75.0 mV
- g_h selected: 0.0005 S/cm2

Key results:

- Sag recovery: control=-0.00 mV, HCN=1.50 mV
- Absolute gain can be lower with HCN because Ih increases membrane conductance
- Normalized impedance: HCN RI > control RI
- HCN peak frequency: 8.0 Hz
- Theta prominence: HCN > control at soma and dendrite

Conclusion:

Under hyperpolarized subthreshold conditions, HCN/Ih produces a relative theta-band resonance. This is a normalized frequency-tuning effect, not an absolute voltage-amplitude amplification.